

Solution Description

This study builds a solid econometric process to predict residential property prices using the Ames Housing dataset within the General Linear Model framework. The goal is to estimate a continuous response variable (SalePrice) as a linear function of various structural and categorical predictors. Ordinary Least Squares (OLS) is chosen as the primary estimation method because it is interpretable and meets the Gauss–Markov assumptions, providing a good balance between predictive performance and statistical inference.

The process starts with careful data preprocessing. This transforms the raw dataset into a structured design matrix suitable for regression. Missing numerical values are filled in using median substitution to protect against outliers. Categorical missing data is treated as a separate level to keep structural information intact. To ensure an unbiased evaluation, the dataset is divided into training, validation, and testing subsets through stratified sampling based on the quantiles of the target variable. This maintains consistent distributions across the splits.

Exploratory data analysis shows that the SalePrice distribution is significantly right-skewed. To address this and meet the normality assumptions required for OLS, a logarithmic transformation is applied to the target variable. This transformation stabilizes variance, reduces skewness, and improves the reliability of hypothesis testing. Correlation analysis identifies key predictors that have strong linear relationships with the target, which guides subsequent feature selection.

Feature engineering is crucial for creating a stable and interpretable model. Categorical variables are transformed into numerical form using dummy encoding, with one category left out to avoid perfect multicollinearity. However, this high-dimensional encoding creates significant multicollinearity among predictors. To handle this, a two-step approach is taken: first, structurally redundant variables are removed (for example, features that are included in aggregate measures), and second, an iterative Variance Inflation Factor (VIF) process is used to remove predictors showing too much linear dependence. This guarantees that the final design matrix is full-rank and numerically sound.

The refined feature set is then used to estimate the OLS model using the closed-form normal equations. The statistical significance of predictors is checked through t-tests, while the overall performance of the model is measured using R^2 , adjusted R^2 , and the F-statistic. To enhance model simplicity and isolate significant signals, predictors with insignificant coefficients ($p > 0.05$) are dropped, and the model is re-estimated using the reduced feature set. This leads to a compact but highly explanatory model that has strong predictive power and improved numerical stability.

Model diagnostics are performed to confirm the assumptions behind OLS. Residual plots show mild heteroscedasticity, which is verified with the Breusch–Pagan test. Although the OLS estimator remains unbiased, the presence of non-constant variance suggests inefficiency in standard errors. This indicates a need for robust inference or regularized methods. Normality checks using Q-Q plots demonstrate acceptable adherence, especially after the logarithmic transformation.

In summary, the final solution provides a high-performing and interpretable regression model with strong explanatory power ($R^2 \approx 0.90$). This is supported by thorough statistical validation and careful feature engineering. While there are minor issues with homoscedasticity, the approach lays a strong groundwork for further refinement using techniques like Ridge or LASSO regression, ensuring both robustness and scalability in practical property valuation tasks.